Solutions to the Exercises of Chapter 12 (Strange Attractors)

S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed.

Contents

Conventions and notes. The Hénon map is $x_{n+1} = 1 - ax_n^2 + y_n,\ y_{n+1} = bx_n$, with Jacobian $J = \begin{pmatrix} -2ax & 1 \\ b & 0 \end{pmatrix}$, $\det J = -b$. The baker's map is $B(x,y) = (2x, ay)$ for $0 \le x < \frac{1}{2}$ and $(2x-1, ay+\frac{1}{2})$ for $\frac{1}{2} \le x \le 1$. The forced double-well oscillator is $\ddot{x} + \delta\dot{x} - x + x^3 = F\cos\omega t$. The text extraction of the book lost some symbols, so problem statements are paraphrased and a few parameter names (δ, ω) are read from context. Computations were done in Python (NumPy/SciPy) with fixed-step RK4 or adaptive DOP853; all figures were generated for this document. Statements that rest on numerics are marked "numerically".

12.1 The Simplest Examples

Exercise 12.1.1

(Uncoupled linear map) $x_{n+1} = ax_n$, $y_{n+1} = by_n$. Draw all patterns of orbits near the origin, depending on the signs and sizes of a, b.

Solution. Since $x_n = a^n x_0$ and $y_n = b^n y_0$, each coordinate evolves independently:

- $|a| < 1$: $x_n \to 0$; $|a| > 1$: $|x_n| \to \infty$; $a = 1$: every point on the x-axis direction is fixed; $a = -1$: x_n flips sign every step (period 2); $a = 0$: x collapses to 0 in one step.
- $a > 0$: x_n keeps its sign (monotone motion); $a < 0$: x_n alternates sign, so the orbit hops between left and right half-planes. Same for b and the y-direction.

Combining: the origin is a *stable node* if $|a|, |b| < 1$, an *unstable node* if $|a|, |b| > 1$, and a *saddle* if one modulus is < 1 and the other > 1. The signs decide the "style": $a, b > 0$ gives orbits that creep monotonically along the curves $|y| = C|x|^{\ln|b|/\ln|a|}$; a negative factor makes the orbit alternate between the two branches of such a curve (a "flip"), and if both are negative the orbit alternates between opposite quadrants. Degenerate cases: $|a| = 1$ or $|b| = 1$ gives lines of fixed points / period-2 points; $a = 0$ or $b = 0$ gives collapse onto an axis in one step. Figure **??** shows representative cases.

Exercise 12.1.2

(Stability criterion) For $x_{n+1} = ax_n + by_n$, $y_{n+1} = cx_n + dy_n$ find conditions on a, b, c, d so that the origin is globally asymptotically stable.

Solution. For a linear map $\mathbf{x}_{n+1} = A\mathbf{x}_n$, $\mathbf{x}_n \to \mathbf{0}$ for all initial conditions iff both eigenvalues satisfy $|\lambda| < 1$. Let $\tau = a + d$, $\Delta = ad - bc$, and $p(\lambda) = \lambda^2 - \tau\lambda + \Delta$. *Claim:*

$$|\lambda_{1,2}| < 1 \iff \Delta < 1, \quad p(1) = 1 - \tau + \Delta > 0, \quad p(-1) = 1 + \tau + \Delta > 0,$$

i.e. $|\tau| < 1 + \Delta < 2$.

Proof. ($\Rightarrow$) If the roots are complex conjugates, $|\lambda|^2 = \Delta < 1$, and $p(\pm 1) > 0$ holds since $p > 0$ on all of $\mathbb{R}$. If they are real and in $(-1, 1)$, then $p(1) = (1 - \lambda_1)(1 - \lambda_2) > 0$, $p(-1) = (1 + \lambda_1)(1 + \lambda_2) > 0$, and $|\Delta| = |\lambda_1\lambda_2| < 1$. ($\Leftarrow$) Complex roots: $|\lambda|^2 = \Delta < 1$. Real roots:

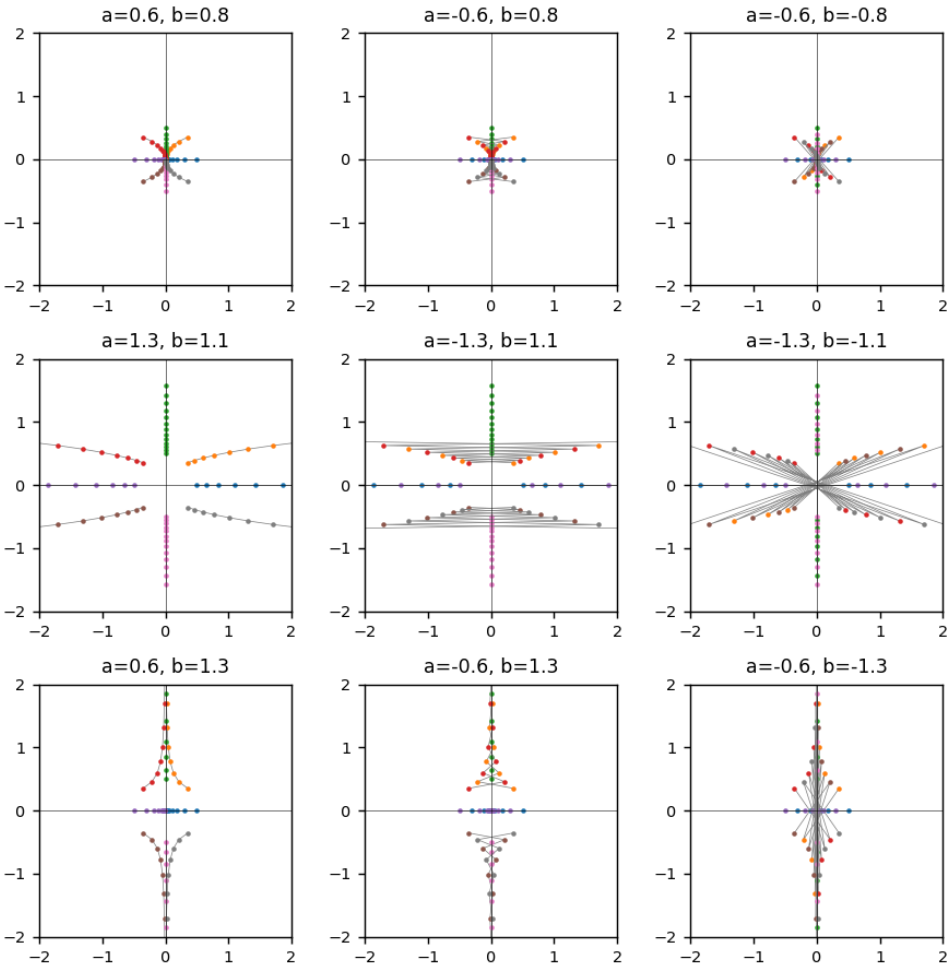


Figure 1: Orbits of the uncoupled linear map for nine representative (a, b): stable nodes (top row), unstable nodes (middle row), saddles (bottom row), with signs $(+,+)$, $(-,+)$, $(-,-)$ in the three columns. Dots are successive iterates joined by thin lines.

$p(1) > 0$ and $p(-1) > 0$ mean that neither 1 nor -1 lies in $[\lambda_1, \lambda_2]$. So either both roots lie in $(-1, 1)$, or both lie above 1 or both below -1 (the roots cannot straddle ± 1 without containing one of them), and in the last two cases $\Delta = \lambda_1\lambda_2 > 1$, excluded by $\Delta < 1$. $\square$

In terms of the original parameters:

$$\boxed{ad - bc < 1, \qquad 1 - (a + d) + (ad - bc) > 0, \qquad 1 + (a + d) + (ad - bc) > 0}$$

This is the triangle in the (τ, Δ)-plane with vertices $(-2, 1), (2, 1), (0, -1)$.

Exercise 12.1.3

Sketch the face of Figure 12.1.4 after one more iteration of the baker's map.

Solution. Figure **??** (with $a = 0.3$) shows the face S, $B(S)$ and $B^2(S)$. The left half of the face ($x < \frac{1}{2}$) is stretched horizontally by 2 and squashed vertically by a onto the lower strip $0 \le y \le a$; the right half goes to the upper strip $\frac{1}{2} \le y \le \frac{1}{2} + a$. Applying B again: each strip is again cut at $x = \frac{1}{2}$, each half is stretched across the full width and squashed by another factor a, and the four resulting strips of height a^2 are stacked at heights 0, $a/2$, $1/2$, $1/2 + a/2$. So the face is now cut into four horizontally stretched, vertically squashed pieces (each is a copy of one quarter of the original, expanded $4\times$ horizontally), and neighbouring features are being pulled apart horizontally.

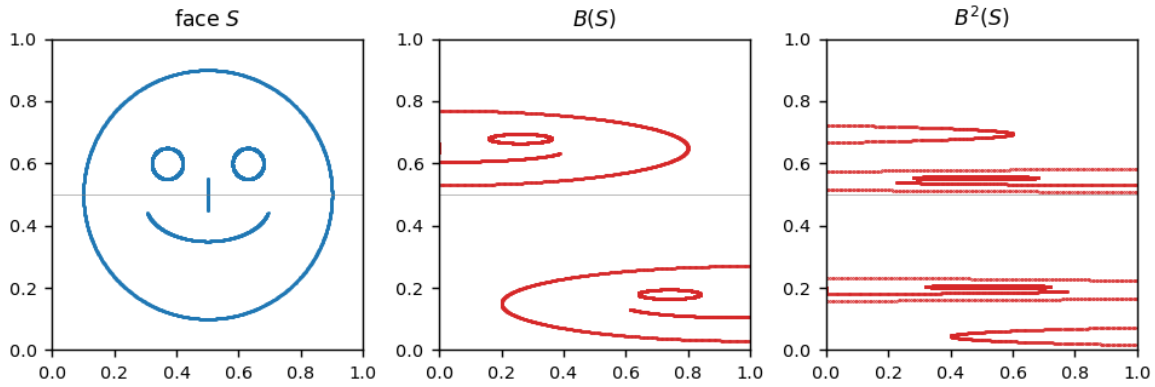


Figure 2: The face, its first image and its second image under the baker's map ($a = 0.3$).

Exercise 12.1.4

(Vertical gaps) Let B be the baker's map with $a < \frac{1}{2}$. (a) Find the largest and smallest gaps in $B^2(S)$. (b) Redo for $B^3(S)$. (c) Answer in general for $B^n(S)$.

Solution. $B^n(S)$ consists of 2^n horizontal strips of height a^n. Each application of B sends a strip with bottom y_0 to two strips with bottoms ay_0 and $ay_0 + \frac{1}{2}$. Hence the bottoms of the strips of $B^n(S)$ are

$$y = \tfrac{1}{2}\sum_{j=0}^{n-1} t_j a^j, \qquad t_j \in \{0,1\}.$$

(a) For $n = 2$ the bottoms are $0,\ \frac{a}{2},\ \frac{1}{2},\ \frac{1}{2}+\frac{a}{2}$ and the strips have height a^2, so the gaps between consecutive strips are

$$\tfrac{a}{2} - a^2, \qquad \tfrac{1}{2} - \tfrac{a}{2} - a^2, \qquad \tfrac{a}{2} - a^2.$$

(The middle gap is larger since $(\frac{1}{2} - \frac{a}{2} - a^2) - (\frac{a}{2} - a^2) = \frac{1}{2} - a > 0$.) Largest gap: $\frac{1}{2} - \frac{a}{2} - a^2$; smallest: $\frac{a}{2} - a^2 = \frac{a}{2}(1-2a)$.

(c) (General n.) Since $a < \frac{1}{2}$ we have $a^j > \sum_{i>j} a^i$, so the order of the strips is the lexicographic order of $(t_0, t_1, \ldots, t_{n-1})$ with t_0 most significant. Two consecutive strips first differ at some index j: the lower one has $t_j = 0, t_{j+1} = \cdots = t_{n-1} = 1$ and the upper one $t_j = 1, t_{j+1} = \cdots = 0$ (equal t_i for $i < j$). The distance between their bottoms is $\frac{1}{2}\big(a^j - \sum_{i=j+1}^{n-1} a^i\big)$, so the gap (distance minus the strip height a^n) is

$$g_j = \tfrac{1}{2}\Big(a^j - \tfrac{a^{j+1}-a^n}{1-a}\Big) - a^n = \frac{1-2a}{2(1-a)}\,\big(a^j - a^n\big), \qquad j = 0, \ldots, n-1,$$

and the gap of size g_j occurs 2^j times (total $\sum_j 2^j = 2^n - 1$ gaps). So

$$\boxed{g_{\max} = g_0 = \frac{(1-2a)(1-a^n)}{2(1-a)}, \qquad g_{\min} = g_{n-1} = \frac{(1-2a)\,a^{n-1}}{2}.}$$

Check $n = 2$: $g_0 = \frac{1}{2}(1-2a)(1+a) = \frac{1}{2} - \frac{a}{2} - a^2$ and $g_1 = \frac{a}{2}(1-2a)$, matching (a).

(b) For $n = 3$: $g_0 = \frac{1}{2}(1-2a)(1+a+a^2)$ (once), $g_1 = \frac{1}{2}a(1-2a)(1+a)$ (twice), $g_2 = \frac{1}{2}a^2(1-2a)$ (four times). The largest is g_0 and the smallest is g_2. (Numerically checked for $a = 0.3, 0.4$.)

Exercise 12.1.5

(Area-preserving baker's map, $a = \frac{1}{2}$) (a) Write the binary representation of $B(x, y)$. (b) Show B has a period-2 orbit and sketch it. (c) Show B has countably many periodic orbits. (d) Uncountably many aperiodic orbits. (e) Are there dense orbits?

Solution. **(a)** Let $x = .a_1a_2a_3\ldots$, $y = .b_1b_2b_3\ldots$ (binary). Then $2x \bmod 1 = .a_2a_3\ldots$, and the new y is $y/2$ if $a_1 = 0$ and $y/2 + \frac{1}{2}$ if $a_1 = 1$, i.e. $y/2$ with leading digit a_1:

$$B(x, y) = \big(.a_2a_3a_4\ldots,\ .a_1b_1b_2b_3\ldots\big).$$

Nicely, if we write the point as the two-sided sequence $\ldots b_3b_2b_1\,.\,a_1a_2a_3\ldots$ (put y reversed next to x), then B simply shifts the binary point one place to the right: $\ldots b_2b_1a_1.a_2a_3\ldots$.

(b) A point has period 2 iff the sequence is invariant under a shift by 2 places, i.e. it is periodic: $\ldots 1010.1010\ldots$ or $\ldots 0101.0101\ldots$ (other 2-periodic sequences are fixed points $\ldots 000.000\ldots = (0,0)$ and $\ldots 111.111\ldots = (1,1)$). These correspond to

$$(x, y) = (\tfrac{2}{3}, \tfrac{1}{3}) \text{ and } (\tfrac{1}{3}, \tfrac{2}{3}),$$

since $.101010\ldots_2 = \frac{2}{3}$ and $.010101\ldots_2 = \frac{1}{3}$. Direct check: $B(\frac{2}{3}, \frac{1}{3}) = (\frac{4}{3} - 1, \frac{1}{6} + \frac{1}{2}) = (\frac{1}{3}, \frac{2}{3})$ and $B(\frac{1}{3}, \frac{2}{3}) = (\frac{2}{3}, \frac{1}{3})$. The orbit consists of these two points, located symmetrically about the centre of the unit square on the anti-diagonal.

(c) A point has period n iff its two-sided sequence is periodic with period dividing n, and such a sequence is determined by n binary digits. So there are at most 2^n points of period n for each n. A countable union of finite sets is countable, so there are countably many periodic points and hence countably many periodic orbits.

(d) Take any x whose binary expansion is not eventually periodic (e.g. any irrational) and any y. The two-sided sequence is then not periodic, so the orbit is aperiodic. There are uncountably many such x (Exercise 11.1.3) and each orbit is countable, so there are uncountably many distinct aperiodic orbits.

(e) Yes. A point has a dense orbit iff its symbol sequence contains every finite block. Let

$$x = .\,0\,1\,00\,01\,10\,11\,000\,001\,010\,011\,100\,101\,110\,111\,0000\ldots$$

(all binary words of length 1, then of length 2, then length 3, …) and take $y = 0$. Given any target (x^*, y^*) and k, we need n so that $B^n(x, y)$ agrees with (x^*, y^*) in the first k binary digits of each coordinate. After n steps $x_n = .a_{n+1}a_{n+2}\ldots$ and $y_n = .a_na_{n-1}\ldots a_{n-k+1}\ldots$, so we need $a_{n-k+1}\ldots a_{n+k}$ to equal the $2k$-digit word formed by the first k digits of y^* reversed followed by the first k digits of x^*. That word appears in the list, and arbitrarily far along. Hence the orbit comes arbitrarily close to every point of the square: it is dense.

Exercise 12.1.6

Study the baker's map on a computer for $a = \frac{1}{2}$. From a random initial condition, plot and label the first ten iterates.

Solution. Figure **??** shows an orbit from a random start $(x_0, y_0) = (0.0856, 0.2368)$. The x-coordinate doubles each step (mod 1), so after a few steps the points jump erratically around the square, while y is obtained by halving and inserting the previous leading digit of x. The iterates look random, as expected from the shift-map description.

Exercise 12.1.7

(Smale horseshoe) (a) Find the points in the original square that map to the two strips. (b) Show that after the next round the square contains four strips; find where they came from. (c) Describe the set of points surviving forever.

Solution. *Setup.* Figure 1 of the book is not reproducible from the text, so I use the standard idealisation. The map is a composition of: stretching the square horizontally by a factor $1/w > 2$ and flattening it vertically by a factor $h < \frac{1}{2}$, folding the resulting long strip into a horseshoe (the two "legs" re-enter the square, the bend hangs over the edge), and cutting off the overhang. Let $L = [0, w] \times [0, 1]$ and $R = [1 - w, 1] \times [0, 1]$ be the vertical strips at the left and right ends of the square.

(a) Points of the middle strip $(w, 1 - w) \times [0, 1]$ are carried into the bend that hangs over the edge, and are lost. The points of L are stretched across the full width and flattened onto a

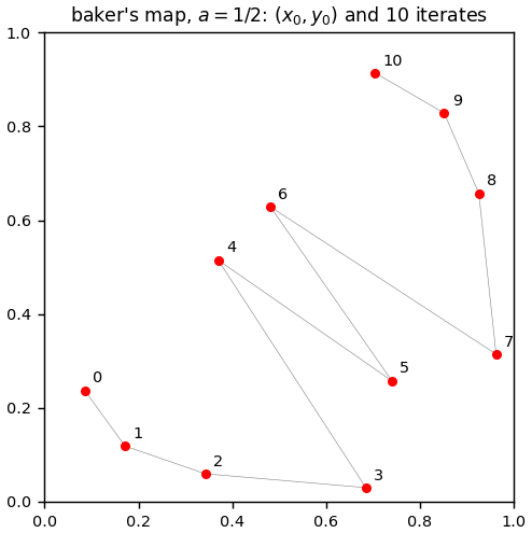


Figure 3: First ten iterates of the baker's map with $a = \frac{1}{2}$, labelled by iteration number.

horizontal strip $H_0 = [0,1] \times [0,h]$ at the bottom; the points of R are carried by the other leg (with the orientation reversed by the fold) onto a horizontal strip $H_1 = [0,1] \times [1-h,1]$ at the top. So the points that survive one iteration are the two *vertical* strips L and R, and they map onto the two horizontal strips H_0, H_1 of the lower-left square in the picture.

(b) A point survives two iterations iff $p \in L \cup R$ and $T(p) \in L \cup R$. Since $T(L) = H_0$ and $T(R) = H_1$ are full-width strips, $T(p) \in L \cup R$ means $T(p)$ lies in one of the four boxes $H_i \cap (L \cup R)$, each of width w and height h. Each such box is the image of a vertical sub-strip of width $w \cdot w = w^2$ inside L or R: explicitly (taking $T|_L : x \mapsto x/w$ and $T|_R : x \mapsto (1-x)/w$)

$$[0, w^2], \quad [w - w^2, w], \quad [1 - w, 1 - w + w^2], \quad [1 - w^2, 1].$$

Their images are four horizontal strips of height h^2 spanning the whole square, lying inside H_0 and H_1 (two in each).

(c) After n iterations the survivors are 2^n vertical strips of width w^n, obtained by removing the middle fraction $1 - 2w$ of each strip at each stage. The set that survives forever is therefore

$$\Lambda_+ = \bigcap_n (\text{survivors after } n \text{ steps}) = C_w \times [0,1],$$

a Cantor set C_w (the symmetric Cantor set that keeps the fraction w at each end, e.g. the middle-thirds set if $w = \frac{1}{3}$) times a vertical segment: a *Cantor set of vertical lines*. It has area $\lim (2w)^n = 0$, so almost every point is eventually mapped into the overhang and escapes (transient chaos). If one asks for orbits that stay in the square for *all* time, forward and backward, one finds also the horizontal Cantor set of lines and the invariant set is $C_w \times C_w$ (Smale's horseshoe), on which the dynamics is conjugate to the shift on two symbols.

Exercise 12.1.8

(Hénon's area-preserving quadratic map) $x_{n+1} = x_n \cos\alpha - (y_n - x_n^2)\sin\alpha$, $y_{n+1} = x_n \sin\alpha + (y_n - x_n^2)\cos\alpha$. *(a) Verify area preservation. (b) Find the inverse. (c) Explore on a computer.*

Solution. **(a)** The Jacobian is

$$\begin{pmatrix} \cos\alpha + 2x\sin\alpha & -\sin\alpha \\ \sin\alpha - 2x\cos\alpha & \cos\alpha \end{pmatrix},$$

with determinant $\cos\alpha(\cos\alpha + 2x\sin\alpha) + \sin\alpha(\sin\alpha - 2x\cos\alpha) = \cos^2\alpha + \sin^2\alpha = 1$. (Equivalently the map is the shear $(x, y) \mapsto (x, y - x^2)$, of determinant 1, followed by a rotation through α.)

(b) Undo the rotation, then the shear. Writing $(x', y') = (x_{n+1}, y_{n+1})$:

$$u = x'\cos\alpha + y'\sin\alpha, \quad v = -x'\sin\alpha + y'\cos\alpha, \qquad x_n = u, \quad y_n = v + u^2,$$

that is $x_n = x'\cos\alpha + y'\sin\alpha$, $y_n = -x'\sin\alpha + y'\cos\alpha + (x'\cos\alpha + y'\sin\alpha)^2$.

(c) Figure **??**: for $\cos\alpha = 0.24$ there is a central elliptic fixed point surrounded by invariant curves, and a chain of five islands around a period-5 cycle (the five points where the five islands are centred). Zooming in on $(-0.57, -0.16)$ reveals smaller island chains and thin chaotic layers, and the structure repeats at finer scales. For $\cos\alpha = 0.22$ the five-island chain is still prominent but now sits in a visible chaotic sea (the dense scatter of points surrounding the islands); orbits starting there wander over a 2-D region rather than along curves. This coexistence of regular islands and chaos is the typical picture for area-preserving maps.

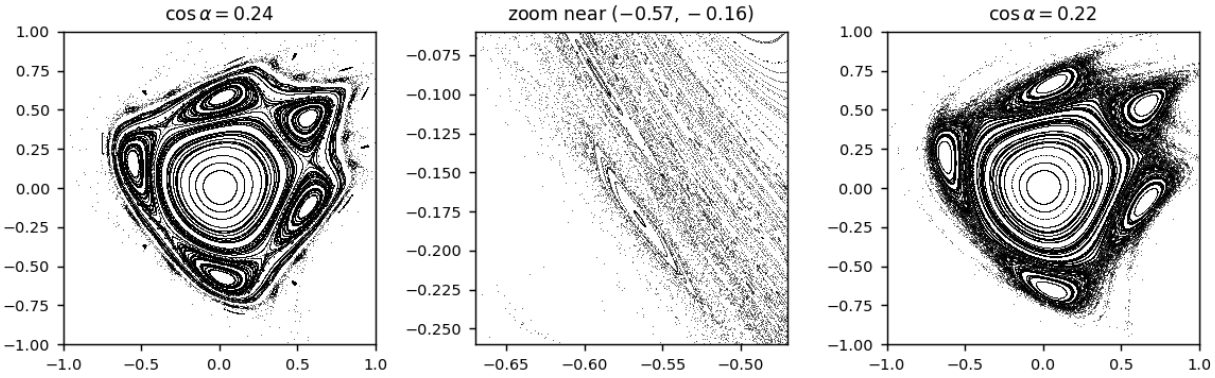


Figure 4: Hénon's area-preserving map: $\cos\alpha = 0.24$ (left), a zoom near $(-0.57, -0.16)$ (middle), and $\cos\alpha = 0.22$ (right). Orbits starting outside the region shown escape and are not plotted.

Exercise 12.1.9

(Standard map) $x_{n+1} = x_n + y_{n+1}$, $y_{n+1} = y_n + k\sin x_n$ (mod 2π). (a) Show area preservation for all k. (b) Plot orbits for $k = 0$. (c) $k = 0.5$. (d) $k = 1$. (e) $k = 2$.

Solution. **(a)** Write $y' = y + k\sin x$ and $x' = x + y' = x + y + k\sin x$. Then

$$\frac{\partial(x', y')}{\partial(x, y)} = \begin{pmatrix} 1 + k\cos x & 1 \\ k\cos x & 1 \end{pmatrix}, \qquad \det = (1 + k\cos x) - k\cos x = 1.$$

Reducing mod 2π does not change derivatives, so the map is area-preserving for all k.

(b) For $k = 0$: y is constant and $x \mapsto x+y$ is a rotation of the circle by y. Orbits lie on horizontal lines; if $y/2\pi$ is rational the orbit is periodic, otherwise it fills the line densely (quasiperiodic). The map is integrable (Figure **??**, $k = 0$).

(c)–**(e)** Figure **??** shows 40 orbits of 2000 iterates each. At $k = 0.5$ almost all orbits still lie on smooth invariant curves (KAM tori), with small islands around resonances (e.g. around $(\pi, 0)$ and $(\pi, 2\pi)$). At $k = 1$ there are both large islands and visible chaotic bands: the thin "fuzzy" regions near the former separatrices. At $k = 2$ the chaotic sea covers most of the square; only the largest remaining islands (the period-1 islands around $(\pi, 0)$) and some small ones survive, so the picture is almost entirely chaotic.

12.2 Hénon Map

Exercise 12.2.1

Show that $T'''T''T'$ is equivalent to formulas (12.2.1).

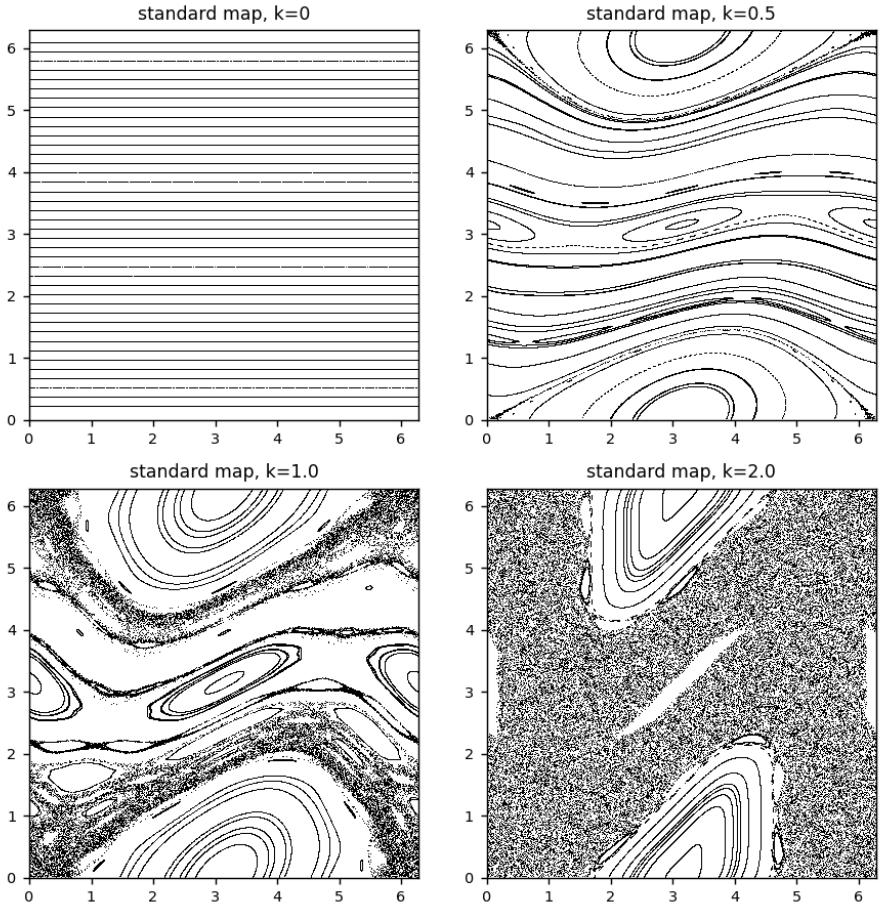


Figure 5: Standard map phase portraits for $k = 0, 0.5, 1, 2$ (40 random initial conditions, 2000 iterates each).

Solution. Start with (x, y). T': $x' = x$, $y' = 1+y-ax^2$. T'': $x'' = bx' = bx$, $y'' = y' = 1+y-ax^2$. T''': $x''' = y'' = 1+y-ax^2$, $y''' = x'' = bx$. So $(x, y) \mapsto (1-ax^2+y,\ bx)$, which is the Hénon map $x_{n+1} = y_n + 1 - ax_n^2$, $y_{n+1} = bx_n$.

Exercise 12.2.2

Show that T' and T''' are area-preserving but T'' is not.

Solution. T': Jacobian $\begin{pmatrix} 1 & 0 \\ -2ax & 1 \end{pmatrix}$, $\det = 1$. T''' (reflection in $y = x$): Jacobian $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\det = -1$, so $|\det| = 1$: area is preserved (orientation is reversed). T'': Jacobian $\begin{pmatrix} b & 0 \\ 0 & 1 \end{pmatrix}$, $\det = b$, with $|b| < 1$, so areas shrink by the factor $|b|$. Hence the Hénon map has $\det J = -b$ and contracts area by $|b|$ at every point.

Exercise 12.2.3

Redo Figure 12.2.1 using an ellipse. (a) Sketch the successive images under T', T'', T'''. (b) Plot accurately with a parametric representation.

Solution. Take the ellipse $x = 1.2\cos\theta$, $y = 0.25\sin\theta$, $a = 1.4$, $b = 0.3$. (a) T' keeps x and replaces y by $1+y-ax^2$: the ellipse is bent into a thin crescent whose upper and lower boundaries are two parabola arcs $y = 1 \pm 0.25\sqrt{1 - x^2/1.44} - ax^2$ (joined at $x = \pm 1.2$). T'' squeezes this horizontally by the factor $b = 0.3$. T''' reflects across $y = x$, so the crescent lies along the x-axis again, with its fold opening to the left. (b) Figure **??** is obtained by pushing the parametrised ellipse through the three maps.

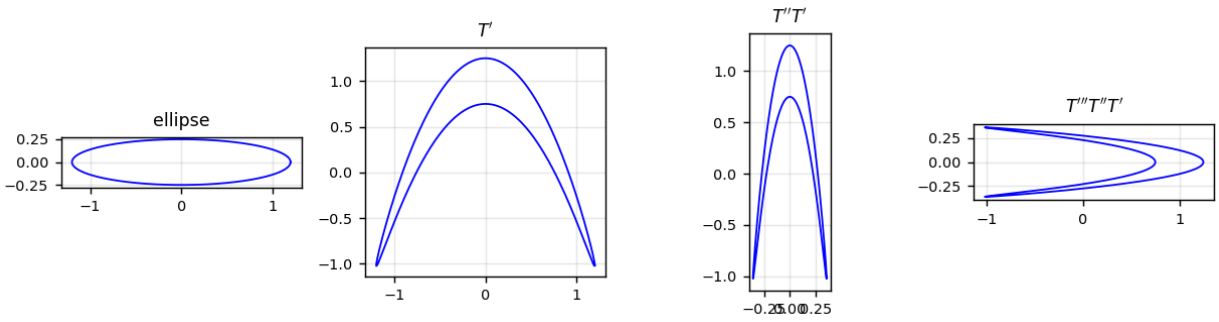


Figure 6: An ellipse and its successive images under T', $T''T'$, $T'''T''T'$ ($a = 1.4$, $b = 0.3$).

Exercise 12.2.4

Find all fixed points of the Hénon map and show they exist only if $a > a_0$.

Solution. A fixed point satisfies $x = 1 - ax^2 + y$, $y = bx$, so $ax^2 + (1-b)x - 1 = 0$ and

$$x^*_{\pm} = \frac{b - 1 \pm \sqrt{(1-b)^2 + 4a}}{2a}, \qquad y^* = bx^* \quad (a \neq 0).$$

Real solutions exist iff $(1-b)^2 + 4a \geq 0$, i.e. $a \geq a_0 := -\frac{1}{4}(1-b)^2$ (for $a = 0$ there is one fixed point $x^* = 1/(1-b)$). Two fixed points exist for $a > a_0$, merging in a saddle-node at $a = a_0$.

Exercise 12.2.5

Compute the Jacobian of the Hénon map and its eigenvalues.

Solution.

$$J = \begin{pmatrix} -2ax & 1 \\ b & 0 \end{pmatrix}, \qquad \det J = -b, \quad \operatorname{tr} J = -2ax.$$

Eigenvalues satisfy $\lambda^2 + 2ax\lambda - b = 0$, so $\lambda_{\pm} = -ax \pm \sqrt{a^2x^2 + b}$. At a fixed point write $s = ax^* = \frac{1}{2}\big(b - 1 \pm \sqrt{D}\big)$, $D = (1-b)^2 + 4a$.

Exercise 12.2.6

(Stability of fixed points) Determine stability as a function of a, b. Show one fixed point is always unstable, the other is stable for a slightly larger than a_0, and loses stability in a flip bifurcation at $a_1 = \frac{3}{4}(1-b)^2$.

Solution. Let $p(\lambda) = \lambda^2 + 2s\lambda - b$ with $s = ax^*$. By the criterion of Exercise 12.1.2 (here $\Delta = -b$, $\tau = -2s$, $|b| < 1$), both $|\lambda| < 1$ iff $p(1) = 1 + 2s - b > 0$ and $p(-1) = 1 - 2s - b > 0$, i.e.

$$|2s| < 1 - b \iff -\tfrac{1}{2}(1-b) < s < \tfrac{1}{2}(1-b).$$

Fixed point with $+$: $s_+ = \frac{1}{2}(b - 1 + \sqrt{D}) > -\frac{1}{2}(1-b)$ always (since $D > 0$). The upper bound $s_+ < \frac{1}{2}(1-b)$ is equivalent to $\sqrt{D} < 2(1-b)$, i.e. $D < 4(1-b)^2$, i.e. $4a < 3(1-b)^2$. So this fixed point is *stable for* $a_0 < a < a_1 = \frac{3}{4}(1-b)^2$. At $a = a_0$ ($D = 0$) we get $s = -\frac{1}{2}(1-b)$, so $p(1) = 0$ and $\lambda = +1$: a saddle-node, so the stable point is born stable just above a_0. At $a = a_1$, $p(-1) = 0$, $\lambda = -1$: a flip (period-doubling) bifurcation. *Fixed point with $-$:* $s_- = \frac{1}{2}(b - 1 - \sqrt{D}) < -\frac{1}{2}(1-b)$, hence $p(1) = 1 + 2s_- - b < 0$, so $\lambda = 1$ lies between the two roots: one eigenvalue exceeds 1. This fixed point is always unstable (a saddle).

Exercise 12.2.7

(2-cycle) For $-1 < b < 1$, show the map has a 2-cycle for $a > a_1$. For which a is it stable?

Solution. Write the map as $x_{n+1} = 1 - ax_n^2 + bx_{n-1}$. A 2-cycle alternates p, q with $p \neq q$:

$$p = 1 - aq^2 + bp, \qquad q = 1 - ap^2 + bq.$$

Subtracting: $(1-b)(p-q) = a(p-q)(p+q)$, so $p+q = S := (1-b)/a$. Adding and using $p^2+q^2 = S^2 - 2pq$ gives $pq = P := \big((1-b)^2 - a\big)/a^2$. Thus p, q are the roots of $z^2 - Sz + P = 0$, real and distinct iff

$$S^2 - 4P = \frac{4a - 3(1-b)^2}{a^2} > 0 \iff a > a_1 = \tfrac{3}{4}(1-b)^2.$$

So the 2-cycle is born at a_1, exactly where the fixed point loses stability (a supercritical flip).

Stability. The cycle points are (p, bq) and (q, bp), and the multipliers are the eigenvalues of $J(q)J(p)$ where $J(x) = \begin{pmatrix} -2ax & 1 \\ b & 0 \end{pmatrix}$:

$$J(q)J(p) = \begin{pmatrix} 4a^2pq + b & -2aq \\ -2abp & b \end{pmatrix}, \qquad \det = b^2, \quad \tau = 4a^2P + 2b = 4(1-b)^2 - 4a + 2b.$$

Stability iff $|\tau| < 1 + b^2$. The inequality $\tau < 1 + b^2$ reduces to $a > a_1$, and $\tau > -(1+b^2)$ reduces to $4a < 5 - 6b + 5b^2$. Hence

$$\boxed{\tfrac{3}{4}(1-b)^2 < a < \tfrac{1}{4}\big(5 - 6b + 5b^2\big)}$$

For $b = 0.3$ this is $0.3675 < a < 0.9125$, in agreement with the numerics of the next exercise.

Exercise 12.2.8

(Numerical experiments, $b = 0.3$) (a) Show period-doubling can occur, leading to onset of chaos at $a \approx 1.06$. (b) Describe the attractor for $a = 1.3$.

Solution. **(a)** Figure **??** (left) is a bifurcation diagram of x versus a. Detecting the period of the attractor numerically (scanning a in steps of 0.0005) gives the period-doubling sequence

period $1 \to 2$	$2 \to 4$	$4 \to 8$	$8 \to 16$	$16 \to 32$	chaos
$a \approx 0.3675$	0.9125	1.026	1.0515	1.057	≈ 1.058

(the first two agree with the analytic values $a_1 = \frac{3}{4}(0.7)^2 = 0.3675$ and $\frac{1}{4}(5 - 1.8 + 0.45) = 0.9125$ from Exercise 12.2.7). The spacings between successive thresholds shrink by factors of about 4.8, 4.5, 4.6, consistent with Feigenbaum's $\delta = 4.669$, and chaos begins at $a \approx 1.06$.

(b) At $a = 1.3$ the attractor is *not* chaotic: a stable **period-7 cycle**, with points approximately $(0.376, 0.207)$, $(1.023, 0.113)$, $(-0.248, 0.307)$, $(1.227, -0.075)$, $(-1.031, 0.368)$, $(-0.014, -0.309)$, $(0.690, -0.004)$. Several initial conditions converge to it (some escape to infinity), and the Lyapunov exponent is $\approx -0.24 < 0$. It lies in one of the narrow periodic windows visible in the right panel of Figure **??** (the white gap near $a \approx 1.3$); for nearby values such as $a = 1.28$ and 1.32 the Lyapunov exponents are $+0.26$ and $+0.31$ (chaos), and for $a = 1.4$ the system has the familiar strange attractor with $\lambda \approx 0.42$.

Exercise 12.2.9

(Invariant set) With $a = 1.4, b = 0.3$ let Q be the quadrilateral with vertices $(-1.33, 0.42), (1.32, 0.133), (1.245, -0.14), (-1.06, -0.5)$. (a) Plot Q and $T(Q)$. (b) Prove $T(Q) \subset Q$.

Solution. **(a)** Parametrise an edge from P to P' by $p(t) = P + t(P' - P)$, $0 \le t \le 1$. Then $T(p(t)) = (1 - ax(t)^2 + y(t),\ bx(t))$ is a parabola arc (quadratic in t in the first coordinate, linear in the second). Figure **??** shows Q (blue) and the four arcs forming $\partial T(Q)$ (red). The vertices go to $(-1.0565, -0.399)$, $(-1.3064, 0.396)$, $(-1.31, 0.3735)$, $(-1.073, -0.318)$.

(b) *Reduction to the boundary.* T is a homeomorphism of the plane ($b \neq 0$), so $T(Q)$ is the closed region bounded by the Jordan curve $T(\partial Q)$. Q is convex, so if the curve $T(\partial Q)$ lies in Q then so does the region it encloses. Thus it suffices to show that each of the four arcs lies inside each of the four half-planes whose intersection is Q.

Checking the inequalities. Let e_j denote the edge from vertex j to vertex $j+1$ ($j = 1, \dots, 4$ in the order listed) and H_j its inside half-plane. For an edge e_i and half-plane H_j, the signed distance

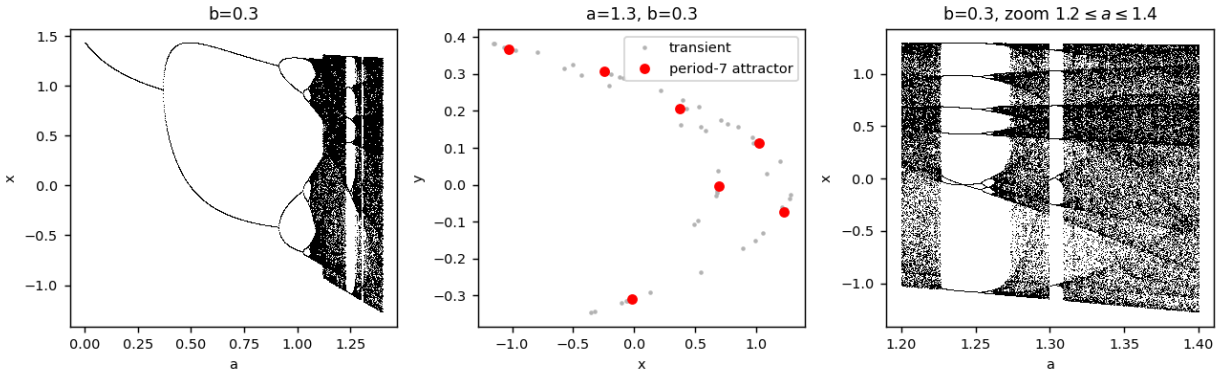


Figure 7: Left: bifurcation diagram for $b = 0.3$. Middle: the stable period-7 cycle at $a = 1.3$ (red), with a transient (grey). Right: magnification of $1.2 \le a \le 1.4$ showing periodic windows inside the chaotic region.

from $T(p(t))$ to the line of e_j is a *quadratic polynomial in* t, so its minimum over $[0, 1]$ is exactly computable (at an endpoint or at the parabola's vertex). The minima, in units of length, are:

arc $T(e_i)\backslash H_j$	H_1	H_2	H_3	H_4
$i = 1$	0.0007	0.0010	0.0201	0.0159
$i = 2$	0.0213	2.5989	0.9016	0.0061
$i = 3$	0.0381	0.5949	0.1419	0.0061
$i = 4$	0.7060	2.1412	0.0992	0.0318

All sixteen minima are strictly positive (I minimised each quadratic on a grid of 20,001 values of t; for a quadratic with these coefficients the grid error is $\sim 10^{-9}$, far smaller than the smallest margin 7×10^{-4}). Hence $T(\partial Q) \subset \operatorname{int} Q$, and therefore $T(Q) \subset Q$. $\blacksquare$ (A random-sample test of 10^5 points of Q also found no point mapped outside Q.) Consequently Q is a trapping region, and the Hénon attractor lies in $\bigcap_{n} T^{n}(Q)$.

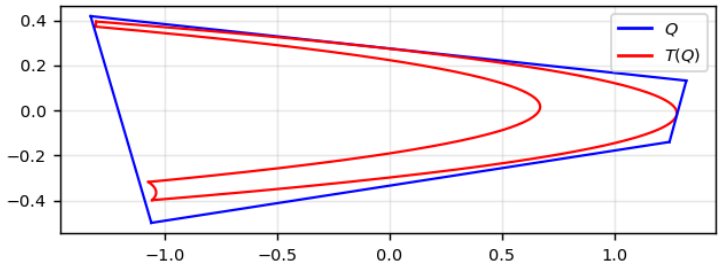


Figure 8: The trapping quadrilateral Q and its image $T(Q)$ for $a = 1.4$, $b = 0.3$.

Exercise 12.2.10

Some orbits of the Hénon map escape to infinity. Find one you can prove diverges.

Solution. Take $a = 1.4$, $b = 0.3$ and start at $(x_0, y_0) = (2, 0)$ (any $|x_0| \ge 2$, $|y_0| \le |x_0|$ works). *Claim:* $|x_n| \ge 2 + 0.6\,n$ for all n and $|y_n| \le |x_n|$. Induction: suppose $r = |x_n| \ge 2$

and $|y_n| \le r$. Since $ar^2 - 1 - r > 0$ for $r \ge 2$ (it is increasing in r and equals 2.6 at $r = 2$), $x_{n+1} = 1 - ax_n^2 + y_n \le 1 - ar^2 + r < 0$ and

$$|x_{n+1}| = ax_n^2 - 1 - y_n \ge ar^2 - r - 1.$$

Write $ar^2 - r - 1 = r + g(r)$ with $g(r) = ar^2 - 2r - 1$. Since $g'(r) = 2ar - 2 > 0$ for $r \ge 2$, $g(r) \ge g(2) = 5.6 - 5 = 0.6$. Hence $|x_{n+1}| \ge r + 0.6$, and $|y_{n+1}| = b|x_n| \le |x_n| \le |x_{n+1}|$. So $|x_n| \ge 2 + 0.6n \to \infty$. Explicitly, $x_1 = -4.6$, $x_2 \approx -28.0$, $x_3 \approx -1.1 \times 10^3$.

Exercise 12.2.11

Show that for a certain choice of parameters the Hénon map reduces to an effectively one-dimensional map.

Solution. Take $b = 0$. Then $y_{n+1} = 0$ for all $n \ge 0$ and $x_{n+1} = 1 - ax_n^2 + y_n$, so after the first step $x_{n+1} = 1 - ax_n^2$: the (quadratic) logistic-type map. The Hénon map is thus a two-dimensional deformation of the logistic family; small $|b|$ gives a "thickened" version of that 1-D map.

Exercise 12.2.12

Suppose we change the sign of b. Is there any difference in the dynamics?

Solution. Yes. The two cases are *not* conjugate in any obvious way; Figure **??** shows that at $a = 1.4$ the map with $b = 0.3$ has the strange attractor ($\lambda \approx +0.42$), while for $b = -0.3$ the attractor is just a stable **2-cycle** at $(0.204, -0.217) \leftrightarrow (0.724, -0.061)$ ($\lambda \approx -0.60$). Differences that can be understood analytically:

- $\det J = -b$ is negative for $b > 0$ (the map reverses orientation) and positive for $b < 0$ (orientation-preserving). For $b > 0$ the eigenvalues $-ax \pm \sqrt{a^2x^2 + b}$ are always real with opposite signs, so orbits near a fixed point alternate sides along the flip direction. For $b < 0$ the eigenvalues have the same sign when real, and become complex (spiralling) when $a^2x^2 < |b|$, in which case $|\lambda|^2 = |b| < 1$.
- The stability criterion of 12.2.6 holds for both signs, so the first flip is again at $a_1 = \frac{3}{4}(1 - b)^2$ (which is 1.2675 for $b = -0.3$, versus 0.3675 for $b = +0.3$), and the 2-cycle is stable for $\frac{3}{4}(1-b)^2 < a < \frac{1}{4}(5 - 6b + 5b^2)$ (here $1.2675 < a < 1.8125$), a much wider window than for $b > 0$. In the bifurcation diagram for $b = -0.3$ (right panel of Figure **??**) the 2-cycle appears at $a \approx 1.27$ as predicted; for the starting point used the orbit escapes to infinity beyond $a \approx 1.72$, so no chaotic attractor is seen in that range.

The area contraction rate $|b|$ is the same.

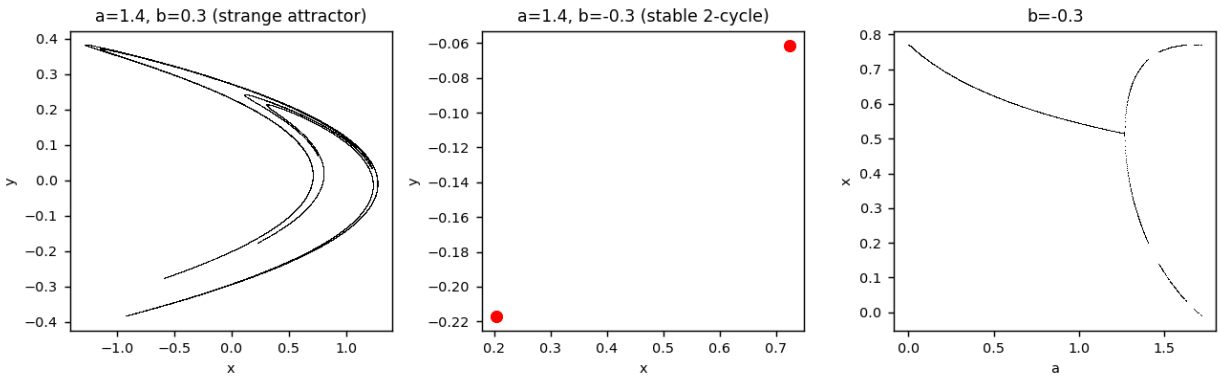


Figure 9: Left: the strange attractor at $a = 1.4$, $b = 0.3$. Middle: the stable 2-cycle at $a = 1.4$, $b = -0.3$. Right: bifurcation diagram for $b = -0.3$.

Exercise 12.2.13

(Computer project) Explore the area-preserving Hénon map ($b = 1$).

Solution. For $b = 1$ we have $\det J = -1$: area-preserving but orientation-reversing; so T^2 is orientation-preserving and area-preserving. Fixed points: $ax^2 = 1$, i.e. $x^* = \pm a^{-1/2}$; eigenvalues $\lambda = -s \pm \sqrt{s^2+1}$ are real and satisfy $\lambda_1\lambda_2 = -1$, so both fixed points are saddles (reflection hyperbolic). Elliptic behaviour therefore comes from period-2 and higher cycles. The orientation-preserving variant $b = -1$ has complex eigenvalues of modulus 1 when $|ax^*| < 1$, i.e. elliptic fixed points. Figure **??** (200 random initial conditions, 1500 iterates) shows the typical Hamiltonian mixture: families of invariant closed curves around elliptic periodic points ($b = 1$: islands at $a = 0.3$; $b = -1$: the large island around the elliptic fixed point), chains of smaller islands, a chaotic sea around them, and orbits escaping to infinity from the outer region. As a increases the regular region shrinks and the chaotic sea grows.

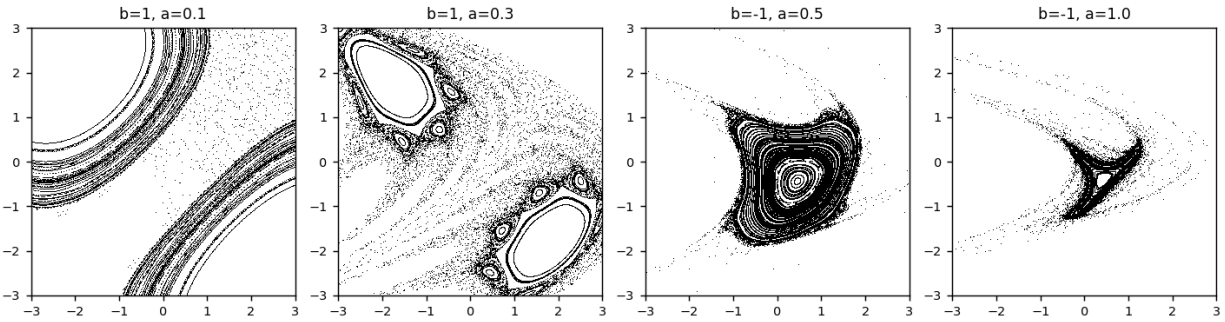


Figure 10: Area-preserving Hénon maps: $b = 1, a = 0.1$; $b = 1, a = 0.3$; $b = -1, a = 0.5$; $b = -1, a = 1.0$.

The Lozi map $x_{n+1} = 1 + y_n - a|x_n|$, $y_{n+1} = bx_n$ (Exercises 12.2.14–12.2.18).

Exercise 12.2.14

In the style of Figure 12.2.1, plot the image of a rectangle under the Lozi map.

Solution. Decompose as for Hénon: T': $(x, y) \mapsto (x, 1 + y - a|x|)$ (a *tent* fold instead of a parabolic fold: the rectangle is bent into an inverted V, with a corner at $x = 0$), T'': $x \mapsto bx$ (horizontal contraction), T''': reflection in $y = x$. Figure **??** shows the images of the rectangle $[-1, 1] \times [-0.1, 0.1]$ for $a = 1.7$, $b = 0.5$. The pieces are straight, with a single kink where the image crosses the fold line $x = 0$.

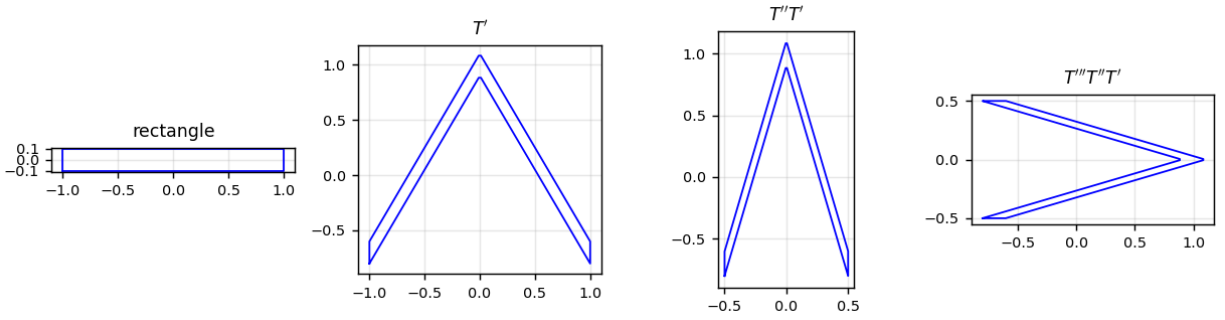


Figure 11: A rectangle and its images under the three stages of the Lozi map ($a = 1.7$, $b = 0.5$).

Exercise 12.2.15

Show the Lozi map contracts areas if $-1 < b < 1$.

Solution. Away from the line $x = 0$ the map is affine on each side, with Jacobian $\begin{pmatrix} -a \operatorname{sgn} x & 1 \\ b & 0 \end{pmatrix}$ and determinant $-b$. So every region is mapped to a region whose area is multiplied by $|b| < 1$

(the line $x = 0$ has zero area, and the map is continuous). Hence areas contract uniformly by the factor $|b|$.

Exercise 12.2.16

Find and classify the fixed points of the Lozi map ($a > 0$).

Solution. Fixed points: $x = 1 + bx - a|x|$, $y = bx$.

- $x > 0$: $(1 - b + a)x = 1$, so $P_+ = \left(\frac{1}{1+a-b}, \frac{b}{1+a-b}\right)$, always present (for $a > 0$, $b < 1$).
- $x < 0$: $(1 - b - a)x = 1$, so $P_- = \left(\frac{1}{1-a-b}, \frac{b}{1-a-b}\right)$, which lies in $x < 0$ iff $a > 1 - b$.

Jacobian at P_+: $\begin{pmatrix} -a & 1 \\ b & 0 \end{pmatrix}$, characteristic polynomial $\lambda^2 + a\lambda - b$ (so $\lambda = \frac{1}{2}(-a \pm \sqrt{a^2 + 4b})$). By 12.1.2: $p(1) = 1 + a - b > 0$ and $p(-1) = 1 - a - b$, so P_+ is stable iff $a < 1 - b$; at $a = 1 - b$ an eigenvalue passes through -1 (flip), and for $a > 1 - b$ it is a saddle ($\lambda_- < -1$, $0 < \lambda_+ < 1$). Jacobian at P_-: $\lambda^2 - a\lambda - b$ with $p(1) = 1 - a - b < 0$ whenever P_- exists, so $\lambda_+ > 1$ while $|\lambda_-| < 1$ ($p(-1) = 1 + a - b > 0$): P_- is always a saddle. So for $a < 1 - b$ only the stable P_+ exists; at $a = 1 - b$ it undergoes a flip and P_- is born; for $a > 1 - b$ both fixed points are saddles. (Complex eigenvalues of P_+ occur for $a^2 + 4b < 0$, which needs $b < 0$.)

Exercise 12.2.17

Find and classify the 2-cycles of the Lozi map.

Solution. Write $x_{n+1} = 1 + bx_{n-1} - a|x_n|$ and look for a cycle p, q with $p = 1 + bp - a|q|$, $q = 1 + bq - a|p|$.

- $p, q > 0$: subtracting gives $(p - q)(1 - b - a) = 0$; $p = q$ is a fixed point (the case $a = 1 - b$ is degenerate).
- $p, q < 0$: $(p - q)(1 - b + a) = 0$, again only fixed points.
- $p > 0 > q$: $(1 - b)p - aq = 1$ and $ap + (1 - b)q = 1$, giving

$$p = \frac{1 - b + a}{(1 - b)^2 + a^2}, \qquad q = \frac{1 - b - a}{(1 - b)^2 + a^2},$$

  with $p > 0$ automatically and $q < 0$ iff $a > 1 - b$.

So a genuine 2-cycle $\{(p, bq), (q, bp)\}$ exists iff $a > 1 - b$, born at the flip of P_+. Its multipliers are the eigenvalues of

$$\begin{pmatrix} a & 1 \\ b & 0 \end{pmatrix}\begin{pmatrix} -a & 1 \\ b & 0 \end{pmatrix} = \begin{pmatrix} b - a^2 & a \\ -ab & b \end{pmatrix}, \qquad \det = b^2, \quad \tau = 2b - a^2.$$

Stability requires $|\tau| < 1 + b^2$. The condition $\tau < 1 + b^2$ is always satisfied ($-a^2 < (1 - b)^2$), and $\tau > -(1 + b^2)$ means $a^2 < (1 + b)^2$. Hence the 2-cycle is stable iff

$$\boxed{1 - b < a < 1 + b}$$

(a nonempty interval only for $b > 0$), and it is a saddle-type cycle for $a > 1 + b$. In particular for $a = 1.7, b = 0.5$ we have $a > 1 + b = 1.5$: the fixed points and the 2-cycle are all unstable, consistent with the strange attractor of the next exercise.

Exercise 12.2.18

Show numerically that the Lozi map has a strange attractor for $a = 1.7$, $b = 0.5$.

Solution. Iterating from $(0.1, 0.1)$ gives an orbit confined to $-1.28 \le x \le 1.34$, $-0.64 \le y \le 0.67$ that never settles (Figure **??**). Evidence for a strange attractor:

- the orbit is bounded and fills out a thin, folded, layered set (zooming in shows that apparently single curves split into several parallel curves, again and again: Cantor-like cross-section);

- since the map contracts area by $|b| = 0.5$, the attractor has zero area;
- the largest Lyapunov exponent, computed by following a tangent vector (renormalised at every step) along an orbit of 2×10^5 iterates, is $\lambda_1 \approx +0.47 > 0$, i.e. sensitive dependence on initial conditions;
- all fixed points and the 2-cycle are unstable (12.2.16–17).

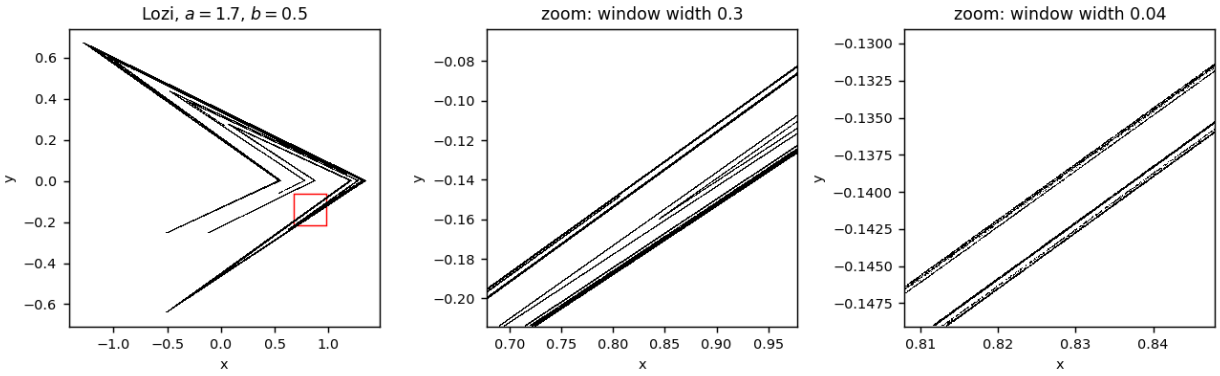


Figure 12: Lozi attractor ($a = 1.7, b = 0.5$) and two successive magnifications of the boxed region (3 million iterates).

12.3 Rössler System

Exercise 12.3.1
(Numerical experiments) $\dot{x} = -y - z$, $\dot{y} = x + ay$, $\dot{z} = b + z(x - c)$ with $b = 2$, $c = 4$; increase a from 0 to 0.4. (a) Find the approximate value of a at the Hopf bifurcation and the first period-doubling. (b) Plot the attractor and the time series $z(t)$ for each a.

Solution. **(a)** *Hopf bifurcation.* The fixed point near the origin is $x^* = \frac{1}{2}(c - \sqrt{c^2 - 4ab})$, $y^* = -x^*/a$, $z^* = x^*/a$ (see 12.3.2). Its characteristic polynomial is $\lambda^3 + A\lambda^2 + B\lambda + C$ with $A = c - x^* - a$, $B = a(x^* - c) + 1 + x^*/a$, $C = c - 2x^*$. A Hopf bifurcation occurs when $AB = C$ (with $A, B, C > 0$), i.e. when a complex pair crosses the imaginary axis. Solving numerically for $b = 2$, $c = 4$ gives

$$a_H \approx 0.125 \quad (x^* \approx 0.063,\ y^* \approx -0.508,\ z^* \approx 0.508).$$

Direct simulation agrees: for $a < 0.125$ orbits spiral into the fixed point; for a just above it they converge to a small limit cycle whose amplitude grows like $\sqrt{a - a_H}$ (so the Hopf bifurcation is supercritical).

Period doubling. Counting the distinct maxima of $x(t)$ on the attractor (long integrations, transients discarded) gives period 1 for $0.13 \le a \le 0.33$, period 2 for $0.335 \le a \le 0.37$, period 4 at $a \approx 0.38$ and period 8 at $a \approx 0.385$, followed by chaos. Bisection locates

$$a_{PD1} \approx 0.3346\ (1 \to 2), \qquad a_{PD2} \approx 0.3746\ (2 \to 4),$$

with the period-doubling cascade accumulating near $a \approx 0.386$–0.39, where the attractor is chaotic. The period of the cycle is about 6.2 throughout.

(b) Figure **??** shows the (x, y) projection and $z(t)$ for $a = 0.2$ (period-1 cycle: one spike of z per loop), 0.34 (period 2: alternating spike heights), 0.38 (period 4) and 0.39 (chaos: irregular spike heights).

Exercise 12.3.2
(Analysis) Find the fixed points of the Rössler system and state when they exist. Classify them. Plot a partial bifurcation diagram of x^* vs c for fixed a, b. Find a trapping region?

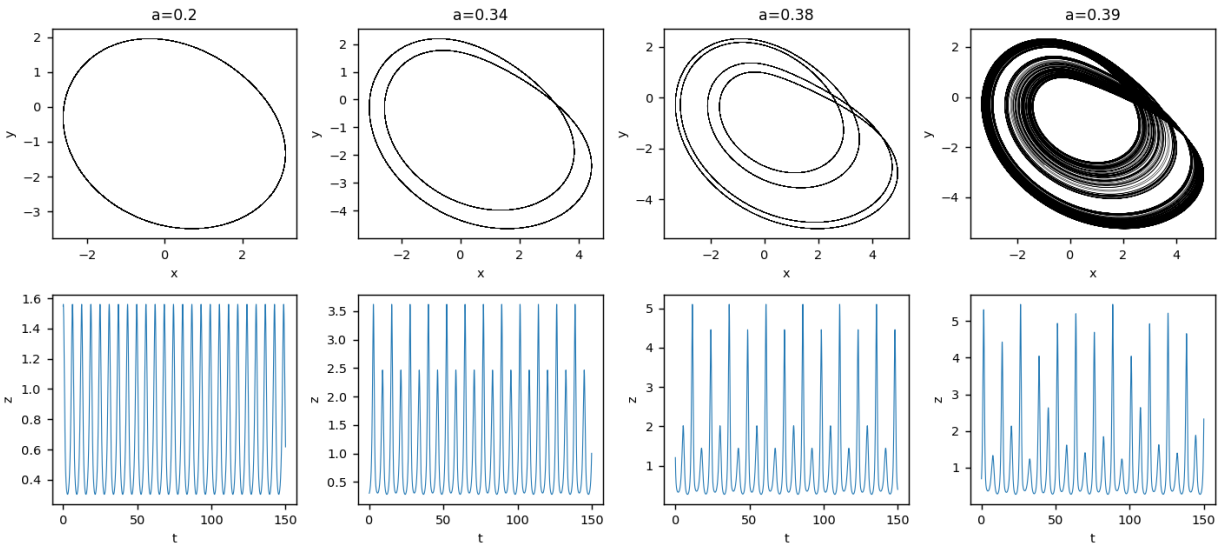


Figure 13: Rössler system with $b = 2$, $c = 4$: attractors in the (x, y)-plane (top) and time series $z(t)$ (bottom) for $a = 0.2, 0.34, 0.38, 0.39$.

Solution. *Fixed points.* $\dot{x} = 0 \Rightarrow z = -y$; $\dot{y} = 0 \Rightarrow x = -ay$; $\dot{z} = 0 \Rightarrow b + z(x - c) = 0$. Substituting, $b + ay^2 + cy = 0$, so

$$y^* = \frac{-c \pm \sqrt{c^2 - 4ab}}{2a}, \qquad x^* = -ay^* = \frac{c \mp \sqrt{c^2 - 4ab}}{2}, \qquad z^* = -y^* = \frac{x^*}{a}.$$

Two fixed points $P_{1,2}$ with $x^*_{1,2} = \frac{1}{2}\big(c \mp \sqrt{c^2 - 4ab}\big)$ exist iff $c^2 \geq 4ab$ (for $a, b > 0$), i.e. $c \geq 2\sqrt{ab}$; they are born in a saddle-node bifurcation at $c = 2\sqrt{ab}$.

Classification. With $e = x^* - c$ the Jacobian is $\begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ z^* & 0 & e \end{pmatrix}$ and its characteristic polynomial is

$$\lambda^3 + A\lambda^2 + B\lambda + C = 0, \quad A = c - x^* - a,\ B = a(x^* - c) + 1 + \tfrac{x^*}{a},\ C = c - 2x^* = \pm\sqrt{c^2 - 4ab}.$$

For P_2 (upper sign choice, $C < 0$): $p(0) = C < 0$ and $p(+\infty) = +\infty$, so there is a positive real eigenvalue. *P_2 is always unstable.* For P_1, $C = \sqrt{c^2 - 4ab} > 0$, and by the Routh–Hurwitz criterion it is stable iff $A > 0$ and $AB > C$; it loses stability through a Hopf bifurcation at $AB = C$. For example, for $a = 0.3$, $b = 0.5$ it is stable for $0.775 < c < 1.133$ and unstable beyond (Figure **??**), whereas for $a = b = 0.2$ (the parameters usually used for chaos, with $c = 5.7$) both fixed points are unstable for every $c \geq 0.4$.

Trapping region. I could not find a simple one, and the usual Lorenz argument fails (see 12.3.3). What can be said rigorously is that the half-space $z \geq 0$ is positively invariant: on $z = 0$ we have $\dot{z} = b > 0$ (for $b > 0$), so orbits cannot leave through that plane. Numerically, for $a = b = 0.2$, $c = 5.7$ the attractor sits in the box $-9.1 \leq x \leq 11.4$, $-10.8 \leq y \leq 7.8$, $0 \leq z \leq 22.9$ (long simulations), but I do not have a proof that a bounded region around it is mapped into itself.

Exercise 12.3.3

The Rössler system has one nonlinear term, yet is harder to analyse than Lorenz, which has two. Why?

Solution. The Lorenz system has a symmetry $(x, y, z) \mapsto (-x, -y, z)$, which halves the analysis and makes the two outer fixed points mirror images; its divergence is the *constant* $-(\sigma + 1 + b) < 0$, so volumes contract at the same rate everywhere; and it has a simple Lyapunov-function argument giving a global trapping ellipsoid (Exercise 9.2.2). The Rössler system lacks the symmetry; its

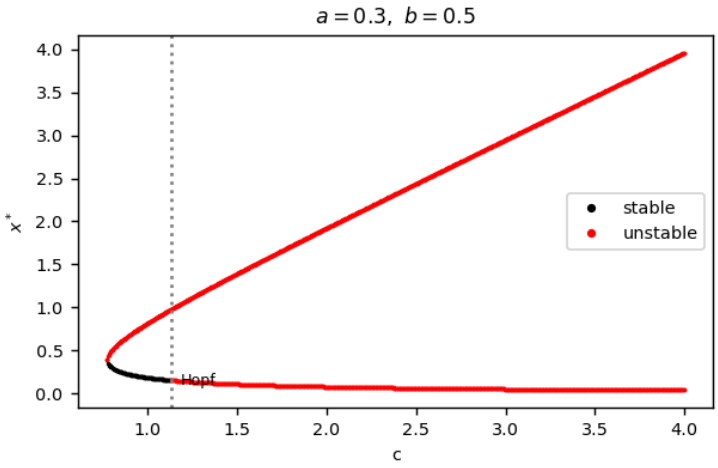


Figure 14: Fixed points x^* of the Rössler system versus c ($a = 0.3, b = 0.5$): stable (black), unstable (red). The pair is born at $c = 2\sqrt{ab} \approx 0.775$; P_1 loses stability in a Hopf bifurcation at $c \approx 1.133$.

divergence $a + x - c$ is positive for $x > c - a$ and negative otherwise, so volumes both expand and contract; its fixed points are given by square-root formulas involving all the parameters, and one of the equations ($\dot{z} = b + z(x - c)$) is exponentially unstable whenever $x > c$, which prevents any simple global bound. So it has fewer terms but less structure.

12.4 Chemical Chaos and Attractor Reconstruction

Exercise 12.4.1

Prove that the time-delayed trajectory in Figure 12.4.5 traces an ellipse for $0 < \tau < \pi/2$.

Solution. Here $x(t) = \sin t$, so the trajectory is $(X, Y) = (\sin t, \sin(t+\tau))$. Introduce the diagonal coordinates $u = (X + Y)/\sqrt{2}$ and $v = (Y - X)/\sqrt{2}$. By the sum/difference formulas,

$$X + Y = 2\sin(t + \tfrac{\tau}{2})\cos\tfrac{\tau}{2}, \qquad Y - X = 2\cos(t + \tfrac{\tau}{2})\sin\tfrac{\tau}{2},$$

so

$$u = \sqrt{2}\cos\tfrac{\tau}{2}\ \sin\big(t + \tfrac{\tau}{2}\big), \qquad v = \sqrt{2}\sin\tfrac{\tau}{2}\ \cos\big(t + \tfrac{\tau}{2}\big).$$

Hence $\dfrac{u^2}{2\cos^2(\tau/2)} + \dfrac{v^2}{2\sin^2(\tau/2)} = 1$: an ellipse with semi-axes $\sqrt{2}\cos\frac{\tau}{2}$ along the diagonal $Y = X$ and $\sqrt{2}\sin\frac{\tau}{2}$ along the anti-diagonal. For $0 < \tau < \pi/2$ we have $\cos\frac{\tau}{2} > \sin\frac{\tau}{2} > 0$, so it is a genuine ellipse with its long axis on the diagonal. (Equivalently in the original variables: $X^2 + Y^2 - 2XY\cos\tau = \sin^2\tau$.) For $\tau = \pi/2$ the semi-axes are equal (a circle), and for $\pi/2 < \tau < \pi$ the long axis lies along $Y = -X$.

Exercise 12.4.2

(Quasiperiodic data) Plot $(x(t), x(t+\tau))$ *for the signal* $x(t) = 3\sin t + \sin(\sqrt{2}\,t)$ *for various* τ. *Does the reconstructed attractor appear to be a torus? Which* τ *seems optimal? Repeat with three-dimensional embeddings.*

Solution. *Note:* the text extraction of the exercise shows $\sin(2t)$, but the exercise is about quasiperiodic data; with frequency ratio 2 the signal would be periodic, so I take the second frequency to be $\sqrt{2}$ (incommensurate with 1), which produces motion on a torus.

Two-dimensional delay plots (Figure **??**). For small τ (0.2) the picture is a narrow band along the diagonal, an almost one-dimensional cigar (since $x(t+\tau) \approx x(t)$). As τ grows the band opens

into an annulus (the projection of a torus into the plane): at $\tau = 1$–1.6 (near a quarter of the dominant period 2π) the ring is widest and the hole is clearest; for larger τ the annulus narrows again and its long axis switches to the anti-diagonal. The plane picture is a *projection* of the torus and so overlaps itself, so it never looks like a surface.

Three-dimensional embedding $(x(t), x(t+\tau), x(t+2\tau))$ (Figure **??**). With $\tau = 1.57 \approx \pi/2$ the trajectory fills a clean torus surface; with $\tau = 0.3$ the torus is squashed into a thin elongated tube along the diagonal. So $\tau \approx \pi/2$, i.e. about a quarter of the dominant period, is a good choice, as in Example 12.4.1: it makes the reconstruction as open as possible, so it is least sensitive to noise.

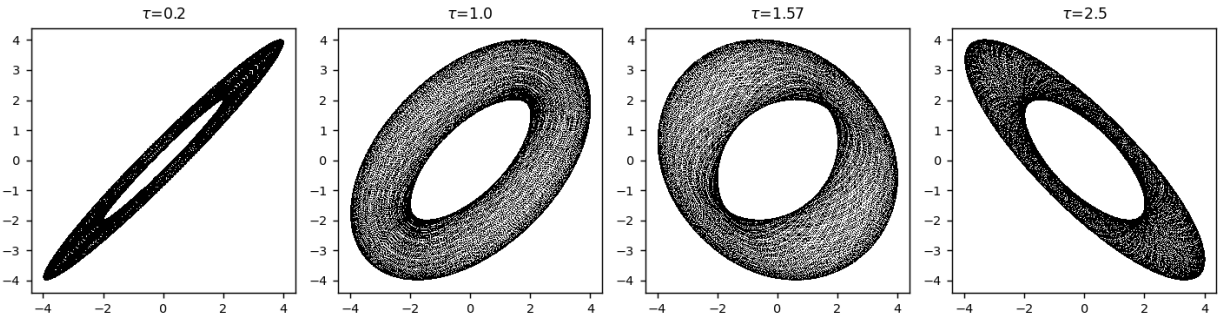


Figure 15: Delay reconstruction $(x(t), x(t+\tau))$ of the quasiperiodic signal for $\tau = 0.2, 1, 1.57, 2.5$.

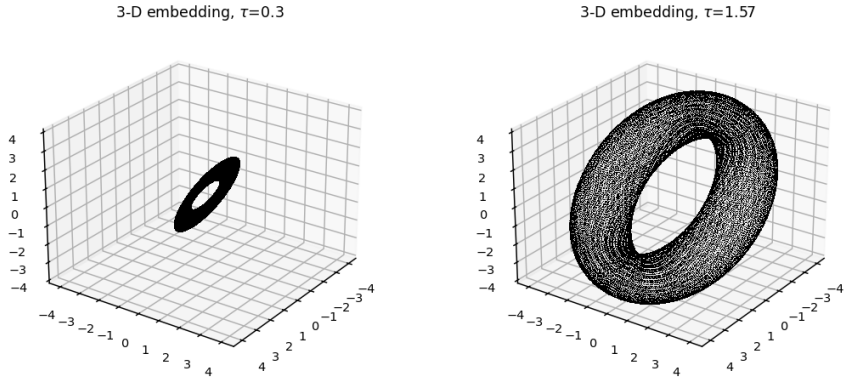


Figure 16: Three-dimensional embedding $(x(t), x(t+\tau), x(t+2\tau))$ for $\tau = 0.3$ (thin) and $\tau = 1.57$ (a clear torus).

Exercise 12.4.3

Integrate the Rössler system for $a = 0.4$, $b = 2$, $c = 4$ and reconstruct the attractor from $x(t)$. Find a τ for which the reconstruction looks like the true attractor. How does it compare with typical orbital periods?

Solution. Figure **??** shows the actual (x, y) attractor and the reconstructions $(x(t), x(t+\tau))$ for $\tau = 0.3, 1.5, 3, 5$. For $\tau = 0.3$ the picture collapses onto the diagonal; for $\tau = 1.5$ it closely resembles the true attractor (a spiral with the same topology: a loop with an inner re-injection and an outer fold); for $\tau = 3$ the picture is a reflected, tilted version with overlaps, and $\tau = 5$ is heavily distorted. The mean time between successive maxima of $x(t)$ (the orbital period) is $T \approx 6.2$, so the good delay $\tau \approx 1.5$ is about $T/4$, within the usual rule of thumb of $T/10$ to $T/2$.

Exercise 12.4.4

Redo for the Lorenz equations with $\sigma = 10$, $r = 28$, $b = 8/3$.

Solution. Reconstructing from $x(t)$ (Figure **??**): $\tau = 0.05$ collapses onto the diagonal (a thickened figure "8"), $\tau = 0.15$ gives a picture with two lobes and the correct topology (two wings with a

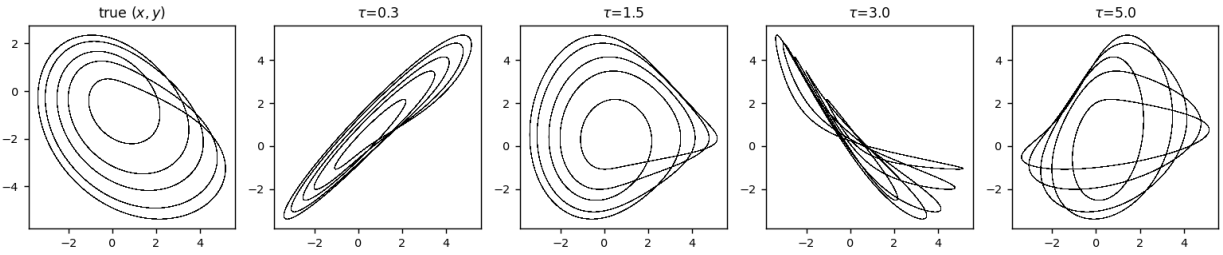


Figure 17: Rössler attractor ($a = 0.4, b = 2, c = 4$): the true (x, y) projection and delay reconstructions for $\tau = 0.3, 1.5, 3.0, 5.0$.

crossover region), similar to the true Lorenz attractor; $\tau = 0.3$ and 0.6 are distorted and overlap. The typical time for one circuit of a wing is ≈ 0.75 (mean time between maxima of z), so the best delay $\tau \approx 0.15$ is about $T/5$, again in the range $T/10$ to $T/2$.

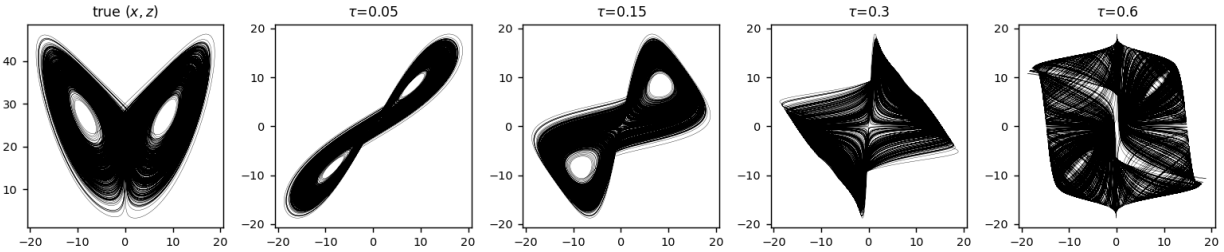


Figure 18: Lorenz attractor: true (x, z) projection and delay reconstructions from $x(t)$ for $\tau = 0.05, 0.15, 0.3, 0.6$.

12.5 Forced Double-Well Oscillator

Exercise 12.5.1

(Basins for the unforced oscillator) Sketch the basins for the weakly damped double-well oscillator in the unforced case $F = 0$. How does their shape depend on the damping? What happens as the damping tends to zero? Implications for predictability?

Solution. For $F = 0$ the system $\ddot{x} + \delta\dot{x} - x + x^3 = 0$ has stable fixed points $(\pm 1, 0)$ and a saddle at the origin. Its energy $E = \frac{1}{2}\dot{x}^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4$ decreases ($\dot{E} = -\delta\dot{x}^2$), so every orbit ends at one of $(\pm 1, 0)$ (once $E < 0$ the orbit is trapped in one well). The basins are separated by the stable manifold of the saddle. Figure **??** shows them: for strong damping ($\delta = 0.5$) the stable manifold winds only a little and the two basins are two large regions; as δ decreases the manifold winds around the wells more often, producing *interleaved spiral bands*, and the bands become thinner (for $\delta = 0.1$ there are already several bands across the picture, and the number of windings scales like $1/\delta$). As $\delta \to 0$ the bands become infinitely thin and infinitely numerous: in the conservative limit there are no attractors at all (orbits are closed curves of constant energy) and the basin partition collapses.

Predictability: the final state (left or right well) of a weakly damped oscillator is extremely sensitive to the initial condition: an uncertainty in the initial state comparable to the band width makes the outcome unpredictable, even though the motion is regular and not chaotic. This is final-state sensitivity.

Exercise 12.5.2

(Coexisting chaos and limit cycle) For $\delta = 0.15$, $F = 0.3$, $\omega = 1$, show numerically that there are at least two coexisting attractors: a large limit cycle and a smaller strange attractor. Plot both in a Poincaré section.

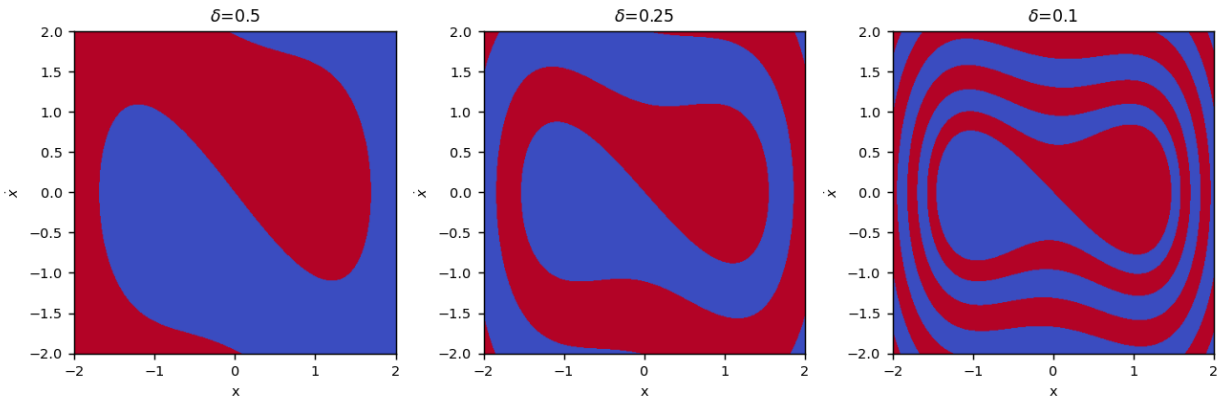


Figure 19: Basins of attraction for the unforced double-well oscillator ($\delta = 0.5, 0.25, 0.1$): red $\to (+1, 0)$, blue $\to (-1, 0)$.

Solution. I integrated 1500 random initial conditions (500 forcing periods of transient) and strobed at $t = 2\pi k$. Two outcomes occurred (Figure **??**):

- about one third of the initial conditions converge to a **period-1 limit cycle** with strobe point $(x, \dot{x}) = (0.639, 1.342)$; during one period x ranges over $[-1.75, 1.75]$, so it is a large-amplitude oscillation that travels back and forth over both wells;
- the others never settle down (their strobe points keep wandering over a set confined to $-1.43 < x < 1.40$, $-0.61 < \dot{x} < 1.13$); a twin-trajectory estimate gives a positive Lyapunov exponent ≈ 0.18 per unit time. This is a **strange attractor**, smaller in extent than the cycle.

Both are stable and coexist, with the long-term behaviour decided by the initial condition.

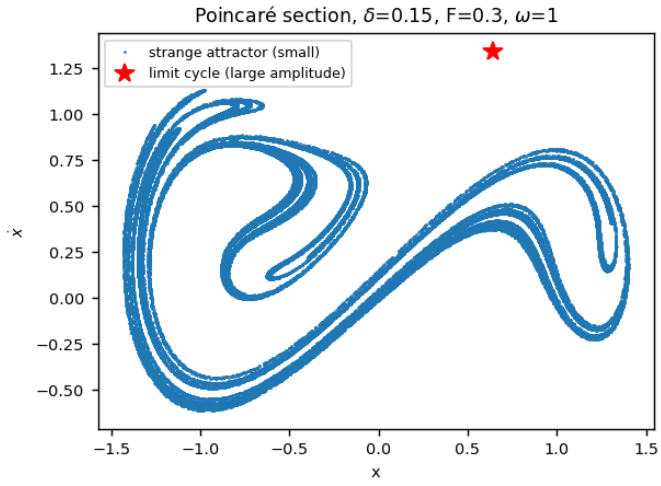


Figure 20: Poincaré section ($t = 2\pi k$): the strange attractor (blue dots) and the large limit cycle (red star), $\delta = 0.15$, $F = 0.3$, $\omega = 1$.

Exercise 12.5.3

(Ueda attractor) $\ddot{x} + k\dot{x} + x^3 = B\cos t$ *with* $k = 0.1$, $B = 12$. *Show numerically that the system has a strange attractor; plot its Poincaré section.*

Solution. Integrating 300 random initial conditions for 200 periods of transient and 100 further strobes each gives the cloud of 3×10^4 points in Figure **??**: a bounded, folded, filamentary set ($0.9 \lesssim x \lesssim 3.95$, $-7.7 \lesssim \dot{x} \lesssim 6.7$ at the strobe times $t = 2\pi k$) that is the same for all initial conditions, with fine layered structure, the signature of a strange attractor.

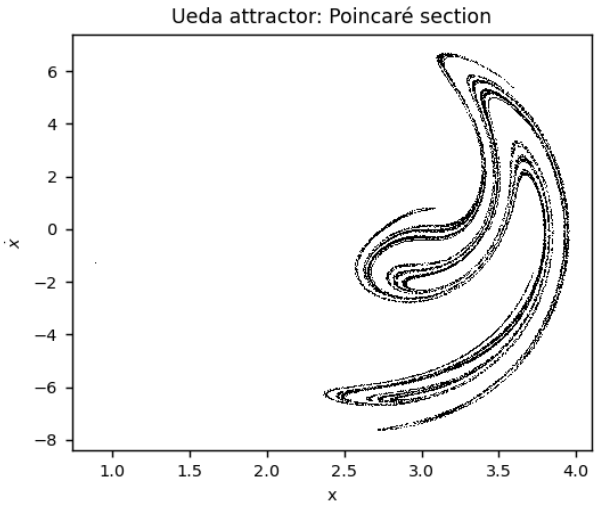


Figure 21: Poincaré section of the Ueda attractor ($k = 0.1$, $B = 12$), strobed at $t = 2\pi k$.

Exercise 12.5.4

(Chaos in the damped driven pendulum) $\ddot{\theta} + b\dot{\theta} + \sin\theta = F\cos t$, $b = 0.22$, $F = 2.7$. *(a) Compute* $\theta(t)$*; interpret. (b) Plot the Poincaré section. (c) Zoom to reveal the Cantor-like cross-section.*

Solution. **(a)** Figure **??** shows $\theta(t)$ (θ not reduced mod 2π). The record is erratic: the pendulum oscillates for a while, then goes over the top and rotates through several revolutions in one direction, then another, then back to oscillations, in an irregular sequence of "tumbles". The sign of the drift changes unpredictably and no repeating pattern is seen.

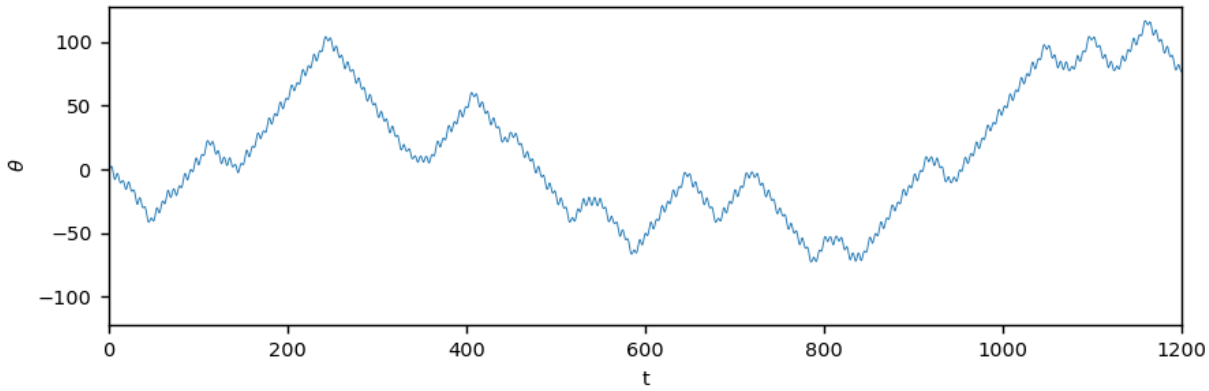


Figure 22: $\theta(t)$ for $b = 0.22$, $F = 2.7$: irregular oscillations and rotations.

(b) Strobing at $t = 2\pi k$ and plotting $(\theta \bmod 2\pi, \dot{\theta})$ for 2.25×10^6 points (1500 initial conditions $\times$ 1500 periods after a transient) gives the folded, layered strange attractor of Figure **??** (left).

(c) Enlarging the boxed region (middle and right panels) shows that what looks like a single curve in the left picture splits into several nearly parallel curves, and each of those again splits when magnified: a Cantor-like cross-section, as for the Hénon attractor.

Exercise 12.5.5

(Fractal basin boundaries) $b = 0.2$, $F = 2$. *(a) Show there are two stable fixed points in the Poincaré section; describe the motion. (b) Compute the basins and show the boundary looks fractal.*

Solution. **(a)** Integrating 400 random initial conditions for 150 forcing periods, all of them

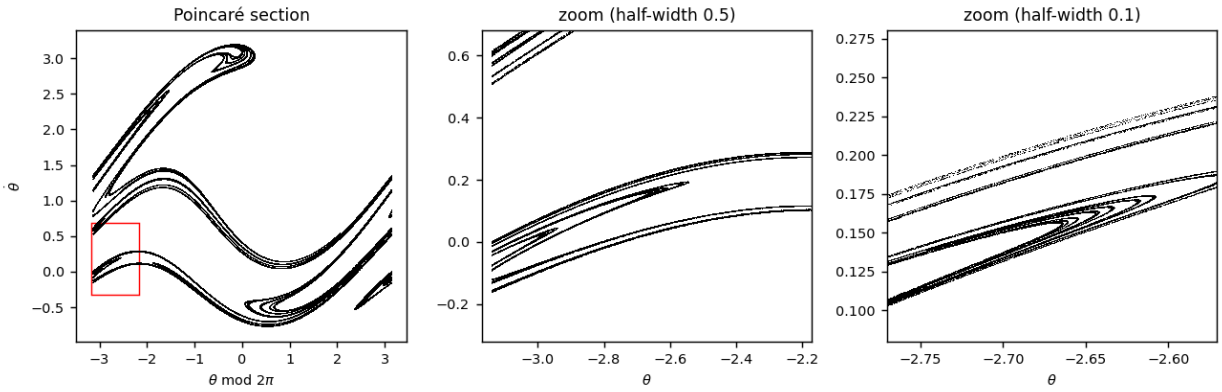


Figure 23: Poincaré section of the pendulum strange attractor and two successive magnifications.

converged to one of two period-1 states (the strobed point repeats exactly, to 10^{-6}):

$$(\theta \bmod 2\pi, \dot\theta) \approx (-0.478,\ -0.608), \qquad \Delta\theta = -2\pi \text{ per period},$$
$$(\theta \bmod 2\pi, \dot\theta) \approx (-0.473,\ \ 2.037), \qquad \Delta\theta = +2\pi \text{ per period}.$$

Each is a *rotating* solution: the pendulum goes once around per forcing period, either in the negative direction or in the positive direction (the average angular velocity is ∓ 1, locked to the drive). The two attractors are mirror-related by the symmetry $(\theta, t) \to (-\theta, t+\pi)$ of the equation.

(b) For each point of a 360×360 grid of initial conditions $(\theta_0, \dot\theta_0) \in [-\pi, \pi] \times [-4, 4]$ I integrated for 70 periods, then classified the point by the direction of rotation ($\Delta\theta = \pm 2\pi$ per period and the matching $\dot\theta$); all but 4×10^{-5} of the points were classified. Figure **??** shows the two basins as intertwined, finely striated regions. Magnifying a boxed region of the boundary (right panel), the same kind of interleaved bands appears on a much smaller scale: the boundary is not a smooth curve but has structure at all resolutions, as expected for a fractal basin boundary. Consequently the final state is very sensitive to the initial condition near the boundary, even though both attractors are simple stable cycles.

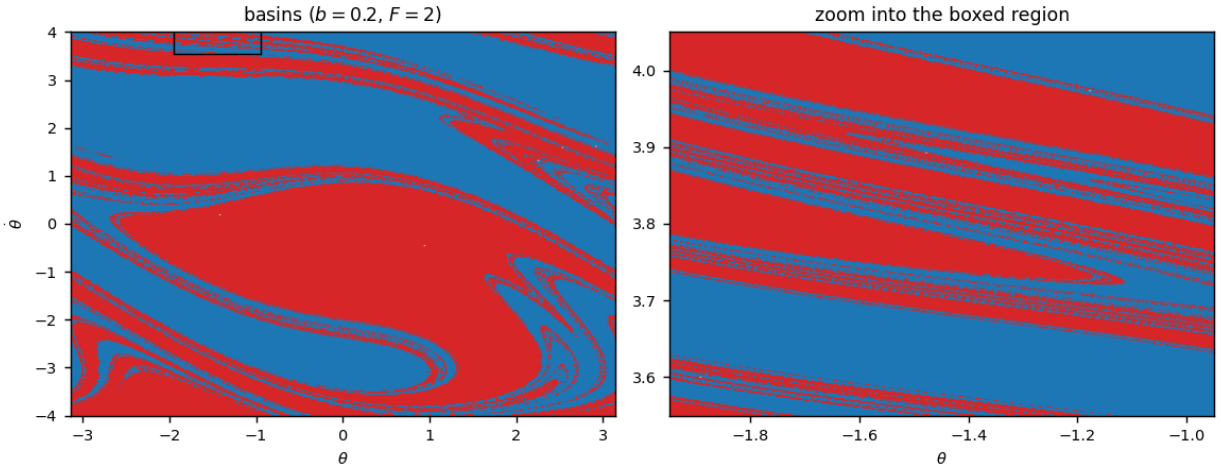


Figure 24: Basins of the two rotating solutions of the pendulum with $b = 0.2$, $F = 2$ (blue: $\Delta\theta = +2\pi$, red: $\Delta\theta = -2\pi$), and a magnification of the boxed region.